D^{2}FT-Net: Frequency–Spatial Dual Domain Fine-Tuning of Vision Foundation Models for Remote Sensing Domain Generalization Semantic Segmentation

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***Abstract*—In recent years, the remote sensing domain generalization semantic segmentation has attracted increasing attention due to significant domain shifts caused by variations in sensors, imaging conditions, and geographic regions. Vision foundation models (VFMs) possess strong general-purpose feature extraction capabilities and can be effectively transferred across diverse data, showing great potential for remote sensing domain generalization and semantic segmentation. However, existing VFM-based spatial-domain parameter-efficient fine-tuning methods struggle to handle pronounced cross-domain intraclass variations in remote sensing. To address this issue, the adaptive frequency-aware adapter (AFA-Adapter) is proposed, which adaptively selects frequency components to improve cross-domain intraclass feature consistency. Building upon this, the spatial multiprototype adapter (SMP-Adapter) is proposed, which clusters multiple prototypes for features of each land-cover category to model complex intraclass diversity. Land-cover features are then weighted by their nearest intraclass prototype, thereby enhancing the discriminability of easily confused features at feature cluster boundaries. By integrating these two modules, we propose a frequency–spatial dual domain fine-tuning network (D^{2}FT-Net), which effectively alleviates cross-domain intraclass variations and improves the generalization capability of VFMs for remote sensing domain generalization semantic segmentation. Extensive experiments under four cross-domain settings demonstrate the effectiveness of the proposed D^{2}FT-Net, which achieves an average mIoU improvement of 1.09% over state-of-the-art methods, with the best gain reaching 1.64%. The source code will be released at https://github.com/ssshen0315/D2FT-Net**

***Index Terms*—Domain generalization, fine-tuning, remote sensing, semantic segmentation, vision foundation models (VFMs).**

Received 26 January 2026; revised 8 April 2026; accepted 3 May 2026. Date of publication 7 May 2026; date of current version 13 May 2026. This work was supported in part by the Fundamental and Interdisciplinary Disciplines Breakthrough Plan of the Ministry of Education of China under Grant JYB2025XDXM115 and in part by the National Natural Science Foundation of China under Grant 42501423. *(Corresponding authors: Xiaoyan Lu; Wenchao Liu.)*

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Digital Object Identifier 10.1109/TGRS.2026.3691111

I. Introduction

REMOTE sensing semantic segmentation is a fundamental task in Earth observation, aiming to assign each pixel of the image to a semantic label [1], [2]. It is widely applied in various fields, including land-cover classification [3], urban planning [4], and environment monitoring [5].

With the rapid advancement of deep learning, various remote sensing semantic segmentation methods have been proposed [7], [8], [9]. However, most existing methods assume that training data and inference data follow independent and identically distributed (i.i.d.) distributions [10]. In practical applications, this assumption often does not hold due to variations in wavelength, geographic region, and spatial resolution. This phenomenon, known as domain shift, poses a significant challenge to deep learning models [11], as models that perform well on training data often suffer significant performance degradation when deployed on unseen inference data [12]. This issue hinders the widespread application of remote sensing semantic segmentation in real-world scenarios [9].

To address this issue, domain adaptation and domain generalization have become two main solutions. Domain adaptation improves performance by aligning source and target distributions using labeled source data and unlabeled target data [13]. However, domain adaptation requires access to target domain data, which is often difficult or expensive to obtain in remote sensing scenarios, thereby limiting its applicability to known target domains [14]. In contrast, domain generalization uses only source domain data for training [15]. It aims to learn domain-invariant features, enabling models to generalize effectively to unseen target domains. As a result, domain generalization alleviates the challenge of limited access to target domain data. Existing domain generalization methods can be categorized into three main strategies: data manipulation, representation learning, and learning strategy [16], [17]. Data manipulation increases robustness by diversifying training data distributions [18]. Representation learning extracts domain-invariant features across domains [19]. Learning strategy enhances generalization by optimizing the training process, such as through meta learning [20]. However, these methods

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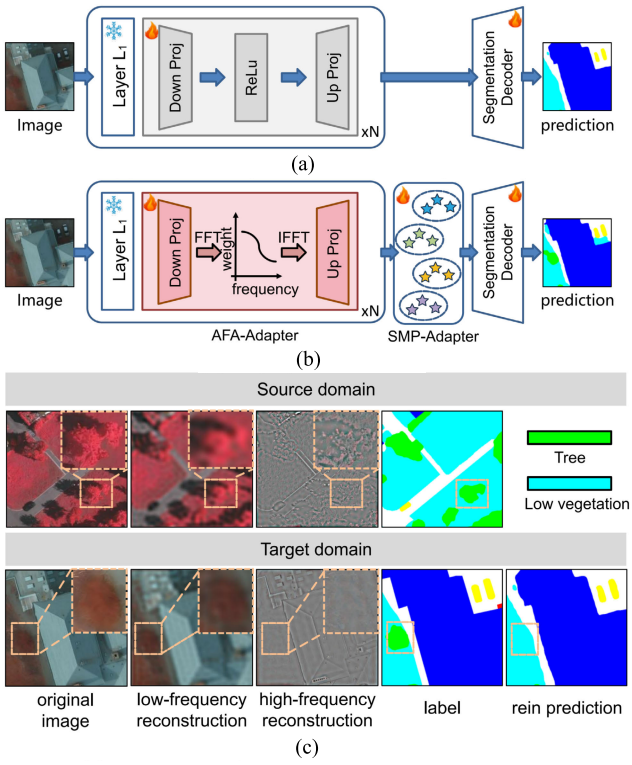


Fig. 1. Conceptual illustrations of (a) existing VFM-based spatial-domain fine-tuning methods. (b) Proposed frequency–spatial D^{2}FT-Net. (c) Qualitative segmentation results of the VFM-based representative method Rein [6] under remote sensing domain generalization. The figure also presents the low-frequency and high-frequency reconstruction of images from both the source and target domains, along with their corresponding labels.

often struggle to achieve sufficient generalization when only limited training data are available [9].

The emergence of vision foundation models (VFMs), including DINOv2 [21] and the segment anything model (SAM) [22], provides new opportunities for domain generalization. By pretraining on large-scale datasets through self-supervised or semi-supervised learning, VFMs acquire general-purpose features that generalize well across diverse data distributions [23]. Therefore, VFMs hold great potential for the remote sensing domain generalization semantic segmentation, as parameter-efficient fine-tuning (PEFT) enables effective generalization even with limited labeled data [8]. As illustrated in Fig. 1(a), most existing VFM-based methods for domain generalization semantic segmentation adopt PEFT methods in the spatial domain [6], [24], [25]. Among them, the representative method Rein [6] introduces learnable land-cover category-aware tokens to progressively refine backbone features across layers, thereby enhancing the generalization capability of VFMs. As shown in Fig. 1(c), due to the large cross-domain differences in tree structure and boundary/texture (e.g., clear crowns and sharp contours in the source domain versus blurred circular regions in the target domain), Rein [6] misclassifies trees as low vegetation. This observation highlights how effectively handling cross-domain intraclass variations remains a critical challenge for VFM-based methods in remote sensing domain generalization and semantic segmentation.

Frequency-domain decomposition provides an alternative perspective by decomposing features from diverse domains into different frequency components that encode discriminative cues such as structures and contours [26]. Thus, we transform images into the frequency domain for analysis, where the spectral radius is defined as the Euclidean distance from the spectrum center in the frequency coordinate. We retain frequency components with a radius smaller than an empirically chosen threshold (set to 20) for reconstruction and then invert them back to the spatial domain to obtain the low-frequency reconstruction; conversely, using components with a radius larger than 20 yields the high-frequency reconstruction. As shown in Fig. 1(c), the low-frequency reconstructions of both the source and target domains exhibit similar large-scale, approximately circular patterns with highly consistent shapes, whereas the high-frequency reconstructions display markedly different texture patterns. These observations suggest that, although categories from the source and target domains exhibit substantial differences in the spatial domain, they are similar in the low-frequency range, indicating low intraclass discrepancy at these frequencies. Therefore, adaptively selecting these frequencies enables VFMs to improve cross-domain intraclass feature consistency, alleviating cross-domain intraclass variations.

Motivated by these findings, the adaptive frequency-aware adapter (AFA-Adapter) is proposed. This module adaptively selects frequency components to learn cross-domain intraclass consistent features, thereby accommodating domain variety. Built upon this module, the spatial multiprototype adapter (SMP-Adapter) is also proposed. The SMP-Adapter models intraclass diversity by clustering multiple prototypes for features of each land-cover category. The land-cover features are then weighted by the nearest intraclass prototype. This approach enhances the discriminability of easily confused features at feature cluster boundaries. Therefore, as shown in Fig. 1(b), integrating these two modules presents a frequency–spatial dual domain fine-tuning network (D^{2}FT-Net). This dual-domain design effectively addresses domain variability, alleviates cross-domain intraclass variations, and improves the generalization capability of VFMs for remote sensing domain generalization semantic segmentation. The main contributions of this work are summarized as follows.

1) To address cross-domain intraclass variations in remote sensing domain generalization semantic segmentation, D^{2}FT-Net is proposed, which consists of two key components: the AFA-Adapter and the SMP-Adapter. The AFA-Adapter aims to learn cross-domain intraclass consistent feature clusters, while the SMP-Adapter aims to enhance the discriminability of easily confused features at feature cluster boundaries.
2) To enhance cross-domain intraclass feature consistency, the proposed AFA-Adapter introduces a learnable adaptive frequency mask to dynamically select frequency components, thereby encouraging the same class features to form compact intraclass feature clusters.
3) To enhance the discriminability of easily confused features at feature cluster boundaries, the SMP-Adapter is

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proposed. It clusters multiple prototypes for features of each land-cover category to model complex intraclass diversity. The land-cover features are then weighted by the nearest intraclass prototype, thereby reducing intraclass distances for ambiguous features at cluster boundaries.

The remainder of this article is organized as follows. Section II reviews related work on domain generalization semantic segmentation. Section III presents a comprehensive description of the proposed D²FT-Net. Section IV reports comparative experimental results on public datasets. Finally, Section V concludes the article.

II. Related Works

A. Domain Generalization Semantic Segmentation

Domain generalization aims to learn domain-invariant representations using only source domain data, thereby enabling models to generalize to unseen remote sensing images. Existing domain generalization methods can be broadly categorized into three groups: data manipulation, representation learning, and learning strategies [16], [17]. DPCL [14] projects test samples toward the source domain and improves representations via multilevel contrastive learning. BlindNet [15] enhances robustness to style shifts by covariance alignment and semantic-consistency contrastive learning. HGFormer [19] performs robust segmentation through hierarchical pixel grouping and multiscale mask classification. CMFormer [27] introduces content-enhanced multiresolution mask attention to better balance content modeling and style robustness. FOSMix [28] improves domain generalization for remote sensing segmentation by mixing style information in the frequency domain while preserving essential semantic frequencies to enhance domain-invariant feature learning.

With the emergence of VFMs, their strong general-purpose feature extraction capability has provided a new perspective for domain generalization. To efficiently adapt VFMs to downstream domain generalization tasks, PEFT has been widely adopted. Rein [6] introduces learnable tokens into the backbone, where each token corresponds to a semantic category and refines feature representations across multiple layers, enabling category-aware modulation with minimal additional parameters. CMPCL [24] improves domain generalization by calibrating cross-domain and inter-class prototype discrepancies through uncertainty-guided and hard-weighted multiprototype contrastive learning. DGTL [29] learns domain-invariant representations from VFMs through text-guided token linking and distribution alignment, thereby improving cross-domain semantic segmentation performance. CDG [30] proposes a novel concept-guided domain generalization approach for semantic segmentation to learn domain-invariant semantic concepts while using the learned semantic concepts to calibrate uncertain predictions for improving the generalization ability. FADA [31] alleviates cross-domain discrepancies by decoupling content and style information in the frequency domain via Haar wavelet decomposition. SET [32] further explores spectral representations by decomposing VFM features into phase and amplitude components and learning style-invariant representations through spectral token-based adaptation. NightAdapter [33] further demonstrates the effectiveness of frequency-aware adaptation by modeling illumination-sensitive and illumination-insensitive frequency components for night-time segmentation.

In the remote sensing domain, generalization semantic segmentation, CrossEarth [34] explores the potential of VFMs through a multitask training strategy tailored for geospatial scenarios. De-GLGAN [35] improves cross-domain semantic segmentation by decomposing features into multiscale high- and low-frequency components to facilitate domain-invariant representation learning. GeoSA-BaSA [9] introduces a geospatial semantic adapter to extract robust multiscale features and a batch style augmenter to reduce style reliance, achieving improved cross-domain generalization across multiple remote sensing datasets. Earth-Adapter [8] is the first remote sensing-specific PEFT method designed to mitigate artifact disturbances for cross-domain semantic segmentation. It proposes a frequency-guided divide-and-conquer strategy that decomposes features via DFT and combines a mixture of adaptors (MoAs) to dynamically fuse multiple frequency adaptations, leading to more robust representations under domain shifts. Freq-Adapter [36] improves sea–land segmentation by introducing a parameter-efficient frequency adapter that leverages frequency-domain representations within foundation models.

In summary, existing studies indicate that PEFT of VFMs holds strong potential for remote sensing domain generalization, with frequency-domain modeling playing a key role in mitigating domain shifts. However, most methods operate in a single domain. In contrast, our method adopts a dual-domain perspective, jointly leveraging frequency and spatial modeling to enhance intraclass consistency and cross-domain generalization.

B. Vision Foundation Models

VFMs are typically pretrained on large-scale datasets using self-supervised or semi-supervised learning paradigms, enabling them to learn general-purpose visual representations that exhibit strong transferability across diverse downstream tasks [37]. Moreover, VFMs have been widely adopted in remote sensing and have achieved remarkable performance [38], [39], [40].

From a training objective perspective, VFMs can be roughly divided into two categories. One category aims to learn generic visual representations through self-supervised or pretext tasks, including models such as DINOv2 [21], DINOv3 [41], and EVA02 [42]. The other category focuses on task-oriented foundation models that are trained with explicit task-specific objectives, with SAM [22] being a representative example for segmentation. DINOv2 [21] is a representative self-supervised VFM trained on a large-scale, high-quality unlabeled dataset comprising 142M images, using a discriminative self-supervised pretraining strategy to learn transferable visual representations. DINOv3 [41] builds upon DINOv2 [21] by scaling both the model size and the training data to learn more powerful and general-purpose visual representations. It is trained on approximately 1.7B images, aiming to achieve

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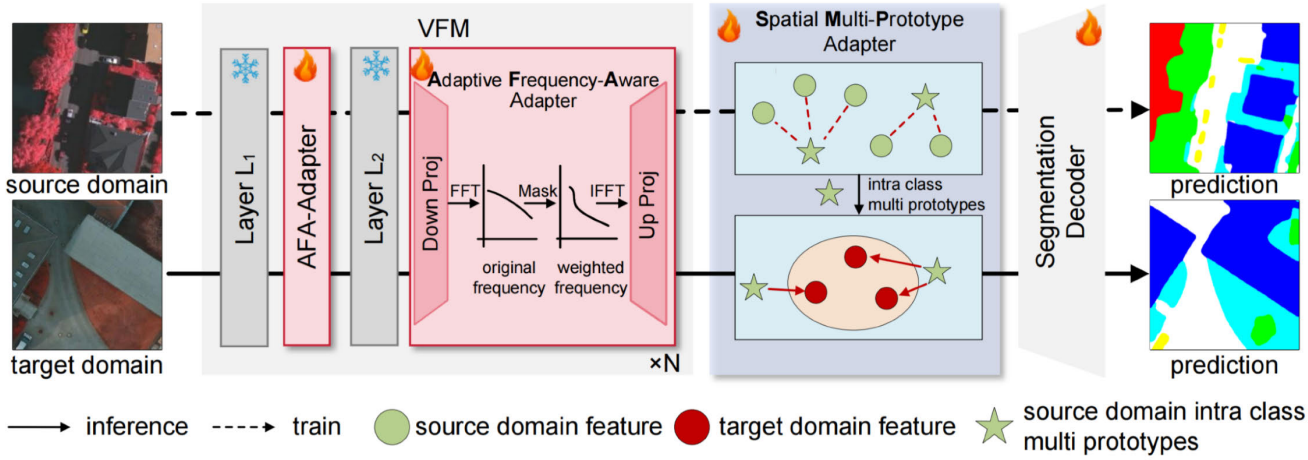


Fig. 2. Pipeline of the proposed D^{2}FT-Net. First, remote sensing images are fed into VFM for feature extraction. The features output from each intermediate layer are fine-tuned by the AFA-Adapter in the frequency domain before being passed to the next backbone layer. After the features from the final backbone layer are processed by the AFA-Adapter, the SMP-Adapter is applied to perform weighted aggregation with the multiprototype features learned during training. The resulting fused features are then fed into the segmentation decoder to produce the final semantic segmentation predictions.

strong downstream performance while keeping the backbone frozen, and ideally reaching SOTA-level results comparable to domain-specific models. EVA02 [42] integrates masked image modeling pretraining with CLIP's vision feature. In contrast, SAM [22] develops a promptable model and pretrains it on a broad dataset for the segmentation task. Notably, the objective of cross-domain generalization is inherently aligned with the zero-shot and transfer learning capabilities of VFMs, suggesting that VFMs hold significant potential for data from diverse sources.

For adapting VFMs to downstream applications, existing studies generally adopt three strategies: full fine-tuning, freezing the backbone and training task-specific heads, or PEFT [43]. Owing to the large parameter scale and rich visual priors embedded in VFMs, PEFT methods have received increasing attention, as they enable efficient model adaptation by updating only a small subset of parameters while significantly reducing computational and memory overhead. Existing PEFT approaches can be broadly categorized into adapter tuning, prompt tuning, and reparameterized tuning [43]. Adapter-based methods insert lightweight learnable modules into the backbone to refine intermediate representations [8]. Prompt tuning introduces learnable prompt tokens at the input stage or across multiple layers to steer the model toward downstream tasks [6]. Reparameterized tuning injects a small number of trainable parameters by reparameterizing weight updates, without directly modifying the original backbone weights [44]. These strategies provide flexible and efficient ways to adapt VFMs with minimal parameter updates, motivating our design to incorporate lightweight adapters for robust cross-domain remote sensing semantic segmentation.

III. Proposed D^{2}FT-Net

A. Architecture of the D^{2}FT-Net

We propose a domain generalization semantic segmentation network for remote sensing images, as illustrated in Fig. 2. The network is built upon a standard remote sensing semantic segmentation architecture, consisting of a pretrained VFM backbone for feature extraction, an encoder, and Mask2Former as the decoder. For a given input image, multilevel features are first extracted by the backbone. During this process, features from each layer are adaptively modulated through the proposed AFA-Adapter module. After the backbone generates high-level features, the SMP-Adapter module is employed to further refine their modulation. The aggregated multilevel features are subsequently fed into the decoder to produce per-pixel segmentation predictions. The network is trained in a supervised manner using a combination of cross-entropy and Dice losses.

D^{2}FT-Net enables PEFT by introducing lightweight adapters in both the frequency and spatial domains to modulate intermediate features. Such a design not only preserves the generalization priors learned by VFMs during large-scale pretraining but also reduces the risk of overfitting to source domain data, leading to stable optimization and high training efficiency. In this way, the proposed network can effectively adapt VFMs to domain-generalized semantic segmentation while maintaining high training efficiency and stable optimization. Specifically, D^{2}FT-Net comprises two key components, namely the AFA-Adapter and the SMP-Adapter, which are applied to different feature levels to jointly alleviate cross-domain intraclass variations.

The AFA-Adapter is inserted into multiple intermediate layers of the VFM to modulate the extracted features. Adaptively selecting frequency components enhances the consistency of intraclass features across domains and encourages features of the same class to form compact intraclass clusters. In contrast, the SMP-Adapter is applied only to the final high-level semantic features output by the backbone network. It models intraclass diversity by clustering multiple prototypes in the spatial domain. Then, pixel-level land-cover features are weighted by their nearest intraclass prototype, which compacts intraclass distances for ambiguous boundary features and further improves discriminability near feature cluster boundaries.

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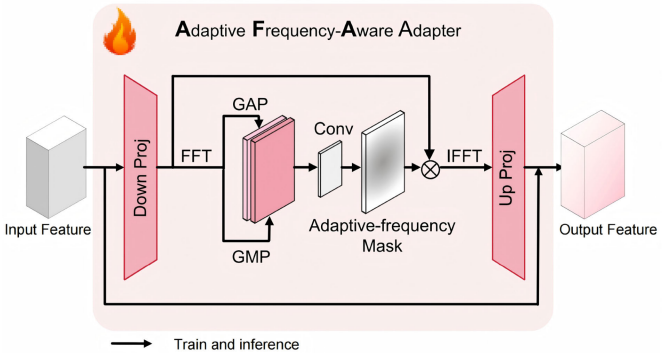


Fig. 3. Architecture of the AFA-Adapter. AFA-Adapter is inserted after each VFM layer to modulate features in the frequency domain via an adaptive frequency mask and then map them back to the spatial domain.

Overall, this dual-domain design effectively alleviates cross-domain intraclass variations: the AFA-Adapter is dedicated to improving the overall consistency of category feature clusters across domains, while the SMP-Adapter enhances the discriminability of easily confused samples near feature cluster boundaries, thereby boosting the generalization capability of VFMs.

B. Adaptive Frequency-Aware Adapter

To enhance cross-domain intraclass feature consistency, the proposed AFA-Adapter introduces a learnable adaptive frequency mask to dynamically select frequency components, thereby encouraging the same class features to form compact intraclass feature clusters.

Different from existing frequency-domain PEFT methods, FADA [31] and Earth-Adapter [8] rely on predefined frequency decomposition strategies to separate high- and low-frequency components, such as Haar wavelet decomposition or threshold-based partitioning. However, such rigid designs limit flexibility. In contrast, the proposed AFA-Adapter introduces a learnable adaptive frequency mask to dynamically select frequency components, enabling more flexible and parameter-efficient frequency adaptation.

The structure of AFA-Adapter is shown in Fig. 3. Given an input features map $\mathbf{f}^n \in \mathbb{R}^{C\times H\times W}$ extracted from the nth layer of the VFM backbone. To reduce computational overhead, the AFA-Adapter first applies a lightweight channel reduction to the input features. The reduced features are then transformed into the frequency domain using a 2-D fast Fourier transform (FFT) applied independently to each channel

$$\mathbf{F}^n = \mathrm{FFT}\left(\mathbf{f}^n\right). \tag{1}$$

To adaptively weight frequency components across different network depths during training, this study introduces a learnable adaptive frequency mask that dynamically reweights frequency components at each layer. The frequency modulation process is formulated as

$$\hat{\mathbf{F}}^n = \mathbf{M}^n \odot \mathbf{F}^n \tag{2}$$

where $\mathbf{F}^n$ denotes the frequency-domain features obtained by FFT, $\mathbf{M}^n$ is a learnable adaptive frequency mask, and $\odot$ indicates element-wise multiplication.

The adaptive frequency mask $\mathbf{M}^n$ is generated based on global frequency statistics. Specifically, we first compute the magnitude spectrum of frequency-domain features $\mathbf{F}^n$ and apply global average pooling (GAP) and global max-pooling (GMP) along the channel dimension to capture complementary frequency responses. The pooled descriptors are then concatenated and passed through a lightweight convolution followed by a sigmoid activation to produce the adaptive frequency mask $\mathbf{M}^n$

$$\mathbf{M}^n = \sigma\left(\mathrm{Conv}\left[\mathrm{GAP}\left(|\mathbf{F}^n|\right), \mathrm{GMP}\left(|\mathbf{F}^n|\right)\right]\right) \tag{3}$$

where $|\cdot|$ denotes the magnitude operation, $[\cdot,\cdot]$ indicates features concatenation, and $\sigma(\cdot)$ is the sigmoid activation function.

After frequency modulation, the filtered features $\hat{\mathbf{F}}^n$ are transformed back into the spatial domain using the inverse FFT (IFFT)

$$\mathbf{z}^n = \mathrm{IFFT}\left(\hat{\mathbf{F}}^n\right). \tag{4}$$

The reconstructed spatial features $\mathbf{z}^n$ are then projected back to the original channel dimension using a lightweight channel expansion layer.

Finally, a residual connection with a learnable scaling parameter a is employed to preserve the original features, yielding the output $\mathbf{f}^n_{\mathrm{out}}$ of the AFA-Adapter

$$\mathbf{f}^n_{\mathrm{out}} = a \cdot \mathbf{f}^n + \mathbf{z}^n \tag{5}$$

where a is a learnable scalar parameter that balances the contribution of the original features and the frequency-weighted features.

The output features $\mathbf{f}^n_{\mathrm{out}}$ are subsequently fed into the next layer of the VFM backbone for forward propagation. By inserting the AFA-Adapter into multiple intermediate layers, the model can adaptively weight different frequency components at different network depths. This design selectively emphasizes frequency components that remain consistent across domains, thereby enhancing cross-domain intraclass feature consistency.

C. Spatial Multiprototype Adapter

To enhance the discriminability of easily confused features at feature cluster boundaries, the SMP-Adapter is proposed. It clusters multiple prototypes for features of each land-cover category to model complex intraclass diversity. The land-cover features are then weighted by the nearest intraclass prototype, thereby reducing intraclass distances for ambiguous features at cluster boundaries.

Among advanced prototype-based domain generalization methods, CMPCL [24] leverages prototypes for contrastive learning to promote domain-invariant features. In contrast, the SMP-Adapter adopts a prototype-guided feature reweighting strategy, which enhances the discriminability of easily confused features at feature cluster boundaries by reducing the distances of ambiguous samples near cluster boundaries.

The structure of the SMP-Adapter is illustrated in Fig. 4. Let $\mathbf{f}^N \in \mathbb{R}^{H\times W\times C}$ denote the last-layer features map extracted by the backbone. We introduce a nonlearnable multiprototype bank $\left\{\mathbf{P}_{k,m} \in \mathbb{R}^C\right\}_{k,m=1}^{K,M}$, where each prototype is determined as the subcluster center of class-specific features. Specifically,

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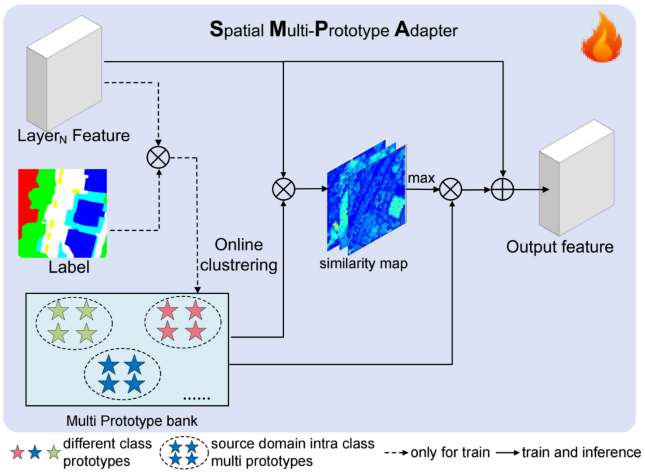


Fig. 4. Architecture of the SMP-Adapter. The input features are first compared with the prototypes stored in the multiprototype bank to compute similarity scores. Based on these scores, the most relevant intraclass prototype is selected to perform weighted feature aggregation. The multiprototype bank is constructed during training via clustering under label supervision.

each semantic class $k \in \{1, \ldots, K\}$ is represented by M prototypes $\{\mathbf{P}_{k,m}\}_{m=1}^{M}$, where $\mathbf{P}_{k,m}$ corresponds to the mth subcluster center of training pixels belonging to class k, thereby enabling effective modeling of intraclass diversity. To compute the similarity between pixel features and prototypes, we flatten $\mathbf{f}^N$ into a set of pixel-wise feature vectors $\{\mathbf{f}_i^N\}_{i=1}^{HW}$ with $\mathbf{f}_i^N \in \mathbb{R}^C$ and compute the cosine similarity between each pixel feature $\mathbf{f}_i^N$ and all prototypes $\mathbf{P}_{k,m}$ as

$$\text{sim}_{i,k,m} = \left\{\left\langle \tilde{\mathbf{f}}_i^N, \tilde{\mathbf{P}}_{k,m} \right\rangle\right\}_{k=1,m=1}^{K,M} \tag{6}$$

where $\tilde{\mathbf{f}}_i^N$ and $\tilde{\mathbf{P}}_{k,m}$ denote the ℓ_2-normalized pixel features $\mathbf{f}_i^N$ and prototypes $\mathbf{P}_{k,m}$, respectively, and $\text{sim}_{i,k,m}$ measures the matching degree between the ith pixel and the mth intraclass prototype of class k

$$(k^*, m^*) = \arg\max_{k,m} \text{sim}_{i,k,m}. \tag{7}$$

For each pixel, we select the most similar prototype and use the selected prototype $\mathbf{P}_{k^*,m^*}$ to enhance the pixel features via weighted additive fusion, thereby compacting the intraclass features distribution and enhancing cross-domain intraclass similarity. This prototype-guided enhancement encourages ambiguous boundary features to move closer to their corresponding intraclass centers, which improves feature compactness and boundary discriminability.

For a multiprototype bank, we adopt an online clustering strategy for prototype assignment and update: guided by the labels, pixel features within the same semantic class are assigned to the corresponding prototype set, and the prototypes are continuously updated according to the assignment results. This clustering and update process encourages the model to explore discriminative subpatterns within each class, and it is worth noting that the multiprototype bank is only updated during training and remains frozen during inference.

IV. Experiments

A. Dataset Description

To validate the proposed method, we use two cross-geographic datasets to conduct our experiments, that is, the ISPRS dataset and the LoveDA dataset.

1) ISPRS Dataset: The ISPRS 2-D semantic labeling dataset [45] consists of two urban aerial image benchmarks, namely the Potsdam dataset and the Vaihingen dataset, which are widely used for cross geographic semantic segmentation and domain generalization evaluation.

a) Potsdam dataset: The Potsdam dataset consists of 38 high-resolution aerial images captured over Potsdam, Germany, with a spatial resolution of 5.0 cm and an image size of 6000 × 6000 pixels. Following the official ISPRS split, 24 images are used for training and 14 for testing. All images are cropped into nonoverlapping 512 × 512 patches, yielding 3456 training and 2016 testing samples. The dataset provides multispectral imagery, and we adopt the IRRG band combination for cross geographic generalization.

b) Vaihingen dataset: The Vaihingen dataset contains 33 high-resolution aerial images acquired over Vaihingen, Germany, with a spatial resolution of 9.0 cm and an approximate image size of 2500 × 2000 pixels. According to the official ISPRS split, 16 images are used for training and 17 for testing. After cropping into 512 × 512 patches, the training and testing sets contain 344 and 398 samples, respectively. Similar to Potsdam, IRRG multispectral imagery is used.

2) LoveDA Dataset: The LoveDA dataset [46] comprises high-resolution images, each with dimensions of 1024 × 1024 pixels and a spatial resolution of 0.3 m, acquired from three Chinese cities. The dataset encompasses two domains: rural and urban scenes, with seven annotated land-cover categories. We follow the official training and validation split [46]. Under this setting, the rural domain contains 1366 training images and 992 validation images, while the urban domain contains 1156 training images and 677 validation images.

B. Experimental Settings

1) Implementation Details: We adopt MMSegmentation for training and evaluation. For the ISPRS Potsdam and Vaihingen datasets, IRRG bands are used as input. Models are optimized using AdamW with layer-wise learning rates, where the decoder and PEFT-related parameters are set to 1×10^{-4}. The batch size is set to 4 for Potsdam and Vaihingen, and 2 for LoveDA, with all models trained for 40 000 iterations. For fair comparison, all competing methods are trained under the same dataset configuration and training protocol.

2) Evaluation Metrics: Following prior work in the remote sensing domain of generalization semantic segmentation, all experiments are conducted with a single run. We adopt IoU and mIoU as the evaluation metrics on the ISPRS and LoveDA datasets. IoU measures the overlap between the prediction and the label

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{8}$$

where TP, TN, FP, and FN are the true positives, true negatives, false positives, and false negatives, respectively.

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TABLE I

QUANTITATIVE COMPARISON WITH REPRESENTATIVE DG METHODS UNDER THE P2V SETTING SOURCE-ONLY DENOTES MODEL TRAINED ON THE SOURCE DOMAIN AND EVALUATED ON THE TARGET DOMAIN WITHOUT ANY GENERALIZATION ORACLE DENOTES MODEL TRAINED AND EVALUATED ON THE TARGET DOMAIN BOLD NUMBERS INDICATE THE BEST PERFORMANCE, AND UNDERLINED NUMBERS DENOTE THE SECOND-BEST RESULTS

Method	Backbone	Publication	Classes						mIoU (%)
			Impervious Surface	Building	Vegetation	Tree	Car	Clutter	
DPCL [14]	ResNet-50 [47]	AAAI 2023	58.8	64.7	41.0	59.4	24.6	5.6	42.34
BlindNet [15]		CVPR 2024	51.7	64.4	25.7	60.4	17.0	4.0	41.80
HGFormer [19]	Swin-T [48]	CVPR 2023	71.57	81.82	46.62	69.89	50.26	35.37	59.25
CMFormer [27]		AAAI 2024	67.85	70.02	37.95	66.43	46.08	7.22	50.59
Source-only	DINOv2 [21]	—	82.39	90.17	61.36	72.92	65.64	43.37	69.31
Oracle		—	87.86	94.39	78.41	80.94	94.23	41.58	79.57
Rein [6]		CVPR 2024	84.08	92.67	64.92	72.38	66.26	43.81	70.68
FADA [31]		NeurIPS 2024	84.58	91.70	65.50	75.93	68.95	40.97	71.27
SET [32]		ACM 2024	83.03	91.30	64.76	78.01	70.02	**45.69**	72.14
CMPCL [24]		T-ITS 2024	85.00	92.56	66.57	74.85	68.17	37.02	70.69
GeoSA-BaSA [9]		ISPRS 2025	**85.63**	92.82	68.73	77.38	68.93	40.53	72.33
Earth-Adapter [8]		AAAI 2026	84.39	92.87	67.21	**78.26**	69.49	37.13	71.56
D²FT-Net (Proposed)		—	84.91	**93.33**	**70.01**	78.12	**70.41**	44.15	**73.49** (+1.16)

The mIoU evaluates the overall segmentation performance by averaging IoU over all categories

$$\text{mIoU} = \frac{1}{N}\sum_{i=1}^{N} \text{IoU}_i \tag{9}$$

where N denotes the number of semantic classes.

3) Comparison Methods: To verify the superiority of the proposed D²FT-Net in domain generalized semantic segmentation for remote sensing images, we compare it with several representative methods, including CNN-based approaches such as DPCL [14] and BlindNet [15], Swin Transformer-based methods such as HGFormer [19] and CMFormer [31], and VFM-based methods including Rein [6], FADA [31], SET [32], CMPCL [24], Earth-Adapter [8], and GeoSA-BaSA [9].

C. Experiments on the ISPRS Dataset

To evaluate cross-geographic generalization on the ISPRS dataset, we conduct experiments under two symmetric settings: Potsdam-to-Vaihingen (P2V) and Vaihingen-to-Potsdam (V2P). In P2V, the model is trained on the Potsdam training set and evaluated on the Vaihingen test set, while in V2P, it is trained on the Vaihingen training set and evaluated on the Potsdam test set. In both settings, no target domain annotations are used during training, ensuring a fair evaluation of cross-eographic generalization.

1) Potsdam-to-Vaihingen:

a) Performance comparison: In this section, comprehensive experiments are conducted under the P2V cross-domain setting to quantitatively compare D²FT with other representative domain generalization methods. The experimental results of different methods are summarized in Table I.

As shown in Table I, VFM-based methods significantly outperform those built upon ResNet-50 and Swin Transformer, demonstrating that introducing VFMs is an effective strategy for improving domain generalized remote sensing semantic segmentation. Building upon this advantage, D²FT achieves the best overall performance, reaching an mIoU of 73.49%, which surpasses the previous state-of-the-art (SOTA) method by 1.16%. Moreover, D²FT ranks among the top two in terms of IoU on five challenging categories, including building, low vegetation, tree, car, and clutter. These categories are typically characterized by significant appearance variations and diverse object scales. The superior performance on these classes demonstrates that the proposed method exhibits stronger robustness and domain generalization capability when handling complex land-cover appearance variations.

To provide a clear performance reference, we follow the commonly adopted cross-domain evaluation protocol [49]. The source-only setting serves as a lower bound, where the baseline model is trained on the source domain and directly evaluated on the target domain without any generalization. The Oracle setting serves as an upper bound, where the baseline model is both trained and evaluated on the target domain. These two settings provide a clear performance range for the cross-domain task and enable a more comprehensive evaluation of model generalization ability.

b) Visualization of the proposed method: To provide a qualitative analysis of the segmentation performance, several representative scenes are selected for visual comparison. Fig. 5 illustrates the qualitative results of D²FT and other representative methods under the P2V cross-domain setting. From left to right, the figure shows the input images, labels, and the predictions generated by different methods.

By comparing the regions highlighted by the dashed boxes in each row, further differences among methods under complex cross-domain scenarios can be observed. Overall, methods based on ResNet-50 and Swin Transformer exhibit inferior visual segmentation quality in cross-domain scenarios, with noticeable misclassification and fragmented predictions. In

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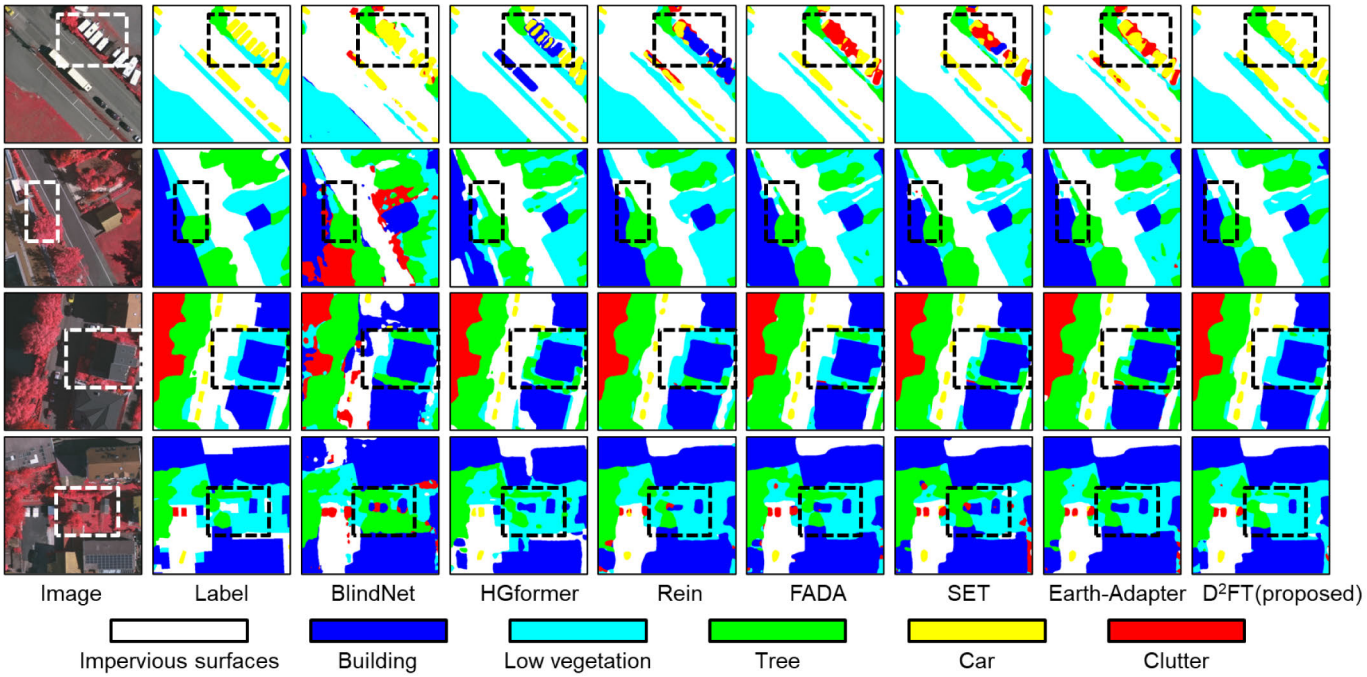


Fig. 5. Visualization results of the proposed D^{2}FT-Net and representative comparison methods on the P2V setting.

TABLE II

QUANTITATIVE COMPARISON WITH REPRESENTATIVE DG METHODS UNDER THE V2P SETTING SOURCE-ONLY DENOTES MODEL TRAINED ON THE SOURCE DOMAIN AND EVALUATED ON THE TARGET DOMAIN WITHOUT ANY GENERALIZATION ORACLE DENOTES MODEL TRAINED AND EVALUATED ON THE TARGET DOMAIN BOLD NUMBERS INDICATE THE BEST PERFORMANCE, AND UNDERLINED NUMBERS DENOTE THE SECOND-BEST RESULTS

Method	**Backbone**	**Publication**	**Classes**						**mIoU (%)**
			Impervious Surface	Building	Vegetation	Tree	Car	Clutter	
DPCL [14]	ResNet-50 [47]	AAAI 2023	67.20	73.40	47.30	28.30	73.40	0.80	48.40
BlindNet [15]		CVPR 2024	66.10	73.70	43.80	25.00	81.10	1.00	48.46
HGFormer [19]	Swin-T [48]	CVPR 2024	66.57	72.80	56.85	36.19	67.77	4.17	50.72
CMFormer [27]		AAAI 2024	64.66	69.42	54.47	26.83	63.70	2.80	47.15
Source-only	DINOv2 [21]	—	76.01	87.71	62.85	46.86	84.76	2.65	60.14
Oracle		—	85.09	93.09	70.31	80.23	73.28	56.7	76.55
Rein [6]		CVPR 2024	<u>77.29</u>	<u>87.84</u>	62.03	48.47	83.19	6.12	<u>60.82</u>
FADA [31]		NeurIPS 2024	76.17	85.00	**64.34**	46.93	79.61	2.03	59.01
SET [32]		ACM 2024	75.80	87.47	62.06	42.74	**86.45**	2.25	59.46
CMPCL [24]		T-ITS 2024	76.19	84.61	64.09	46.90	84.51	5.89	60.36
GeoSA-BaSA [9]		ISPRS 2025	72.80	82.17	62.05	43.19	<u>84.87</u>	**13.44**	60.59
Earth-Adapter [8]		AAAI 2026	75.47	85.30	<u>64.27</u>	**51.31**	80.62	<u>6.27</u>	60.54
D^{2}FT-Net (Proposed)		–	**79.22**	**89.43**	63.41	<u>49.62</u>	81.10	9.83	**61.97** (+1.15)

contrast, VFM-based fine-tuning methods produce more coherent and reliable results.

In the first-row example, several competing methods misclassify a car as building or clutter, whereas D^{2}FT-Net accurately identifies vehicle regions. In the second and third rows, most methods suffer from severe confusion between low vegetation and trees, while D^{2}FT-Net effectively alleviates this issue and generates more consistent segmentation maps. In the fourth-row example, other methods misclassify impervious surfaces as building, whereas our method correctly segments this region. In contrast, D^{2}FT produces more promising segmentation results, intuitively demonstrating its competitive performance in the remote sensing domain generalization semantic segmentation.

2) Vaihingen-to-Potsdam:

a) Performance comparison: In this section, comprehensive experiments are conducted under the V2P cross-domain setting to quantitatively compare D^{2}FT-Net with other representative domain generalization methods. The experimental results of different methods are summarized in Table II.

As shown in Table II, the proposed method achieves an overall mIoU of 61.98%, surpassing the SOTA method by 1.15% mIoU. It also ranks among the top two results on the impervious surface, building, and tree classes. By effectively

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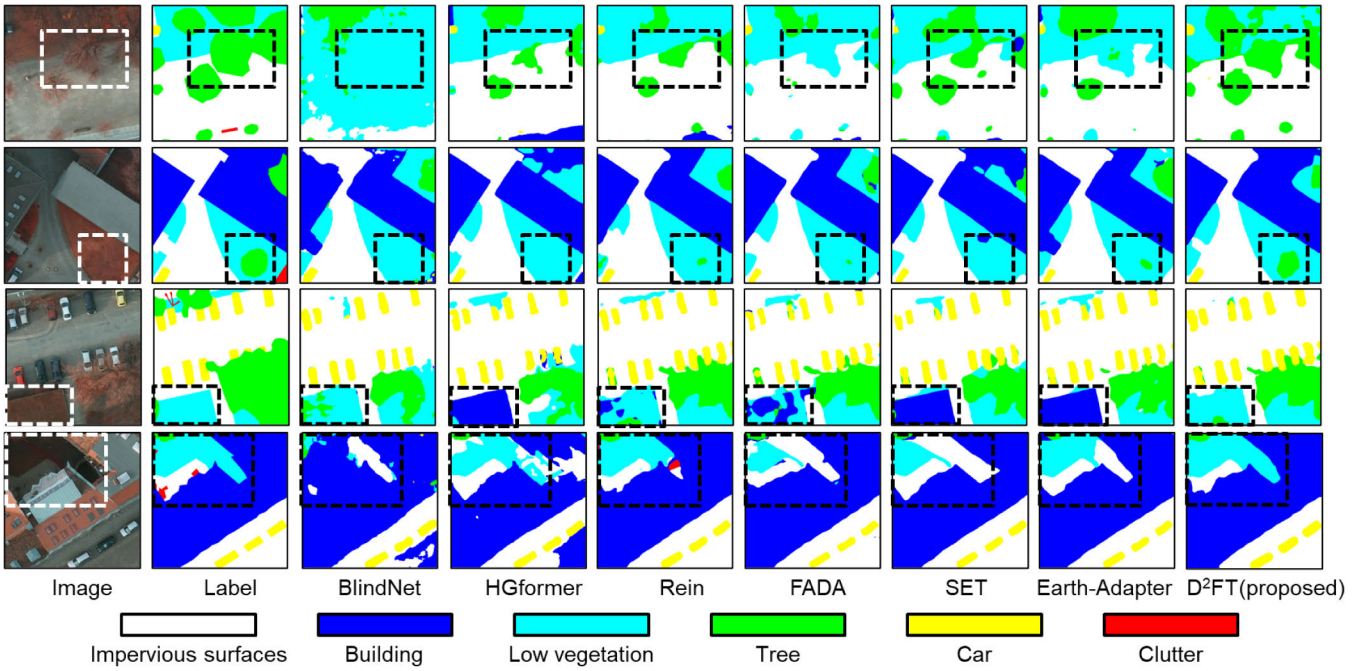


Fig. 6. Visualization results of the proposed D²FT-Net and representative comparison methods on the V2P setting.

mitigating cross-domain intraclass variations, the proposed framework enhances the generalization capability of VFMs in complex remote sensing scenarios, thereby demonstrating its effectiveness for domain generalization semantic segmentation.

b) Visualization of the proposed method: To provide a qualitative analysis of the segmentation performance, several representative scenes are selected for visual comparison. Fig. 6 illustrates the qualitative results of D²FT-Net and other representative methods under the V2P cross-domain setting. From left to right, the figure shows the input images, labels, and the predictions generated by different methods.

By comparing the key regions highlighted by the dashed boxes in each row, further differences among methods under complex cross-domain scenarios can be observed. In the first row, most competing methods tend to misclassify trees as low vegetation, whereas the proposed method delineates tree boundaries more accurately, demonstrating stronger category discriminability. In the second row, the other methods fail to correctly recognize the tree class and suffer from noticeable misclassification, while our method accurately segments the tree regions. In the third row, most methods incorrectly predict low vegetation as building, whereas our method preserves correct classification and effectively alleviates category confusion. In the fourth row, the competing methods further misclassify low vegetation as impervious surface, while the proposed method consistently identifies the true category, exhibiting stronger cross-domain robustness and generalization capability.

3) Ablation Experiments:

a) Complexity comparison: As shown in Table III, we compare D²FT-Net with other VFM-based fine-tuning methods in terms of trainable parameters. The reported parameters only account for those introduced by the fine-tuning modules, excluding the decoder, and all methods adopt the same Mask2Former decoder to ensure a fair comparison. The baseline corresponds to a frozen DINOv2 backbone coupled with a Mask2Former decoder.

TABLE III

COMPARISON OF DIFFERENT METHODS ON THE DINOV2 BACKBONE ON THE ISPRS DATASET. "PARAMS" DENOTES THE NUMBER OF TRAINABLE BACKBONE PARAMETERS

Method	Params (M)	FLOPs (G)	ΔFLOPs (G)	Train (s/iter)	Infer (s/iter)
baseline	0	665.6	0.0	0.58	0.106
Rein [6]	3.00	723.4	57.8	**0.82**	0.112
FADA [31]	**2.06**	701.5	35.9	1.24	0.124
SET [32]	7.54	779.9	114.4	1.36	0.140
CMPCL [24]	2.99	723.4	57.8	1.70	0.130
GeoSA-BaSA [9]	16.69	798.0	132.4	3.94	0.160
Earth-Adapter [8]	9.59	**669.2**	**3.7**	1.18	**0.100**
D²FT-Net	4.22	672.1	6.5	1.10	0.109

Beyond achieving competitive segmentation accuracy, Table III further demonstrates that the proposed D²FT-Net attains a favorable tradeoff between performance and computational efficiency. Specifically, it significantly improves domain generalization segmentation performance while introducing only a marginal number of additional trainable parameters.

b) Performance on different backbones: Table IV reports the generalization performance of D²FT-Net across different VFMs under the P2V setting. As shown in the table, D²FT-Net consistently improves the segmentation performance over the corresponding frozen baselines on all evaluated backbones, including SAM [22], EVA02 [42], DINOv2 [21], and DINOv3 [41], demonstrating that the proposed method is not restricted to a specific backbone architecture. Overall,

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TABLE IV

GENERALIZATION PERFORMANCE OF D^{2}FT-NET ACROSS DIFFERENT VFM BACKBONES UNDER THE P2V SETTING. "PARAMS" DENOTES THE NUMBER OF TRAINABLE BACKBONE PARAMETERS

Backbone	Method	Params (M)	mIoU (%)
SAM [22] (Huge)	Freeze	0.0	56.67
	D^{2}FT-Net	2.96	**59.15**
EVA02 [42] (Large)	Freeze	0.0	60.13
	D^{2}FT-Net	4.22	**67.96**
DINOv2 [21] (Large)	Freeze	0.0	69.31
	D^{2}FT-Net	4.22	**73.49**
DINOv3 [41] (Large)	Freeze	0.0	71.33
	D^{2}FT-Net	4.22	**72.72**

TABLE V

ABLATION STUDY OF THE COMPONENTS IN D^{2}FT-NET UNDER THE P2V SETTING

AFA-Adapter	SMP-Adapter	mIoU (%)	ΔmIoU(%)
×	×	69.31	0.0
✓	×	71.65	+2.34
×	✓	70.47	+1.16
✓	✓	**73.49**	**+4.18**

TABLE VI

COMPARISON OF AFA-ADAPTER WITH OTHER PEFT METHODS

Method	P2V (%)	V2P (%)	Avg (%)
VPT [50]	69.55	59.78	64.67
AdaptFormer [51]	72.04	59.45	65.75
Earth-Adapter [8]	71.04	60.07	65.56
AFA-Adapter	**73.49**	**61.97**	**67.73**

the results indicate that D^{2}FT-Net effectively fine-tunes VFMs, allowing them to achieve improved adaptability and generalization performance in domain generalization remote sensing semantic segmentation tasks.

c) Component effectiveness ablation: Table V presents the ablation results of the individual components in D^{2}FT-Net under the P2V setting. The baseline employs a frozen DINOv2 backbone together with a trainable Mask2Former segmentation head. Incorporating the AFA-Adapter consistently improves baseline performance, raising the mIoU from 69.31% to 71.65%, which demonstrates the effectiveness of frequency-aware adaptation for domain-generalization remote sensing semantic segmentation. By introducing the SMP-Adapter on top of the baseline, the performance improves from 69.31% to 70.47%, demonstrating that the SMP-Adapter alleviates cross-domain intraclass variations. When both the AFA-Adapter and SMP-Adapter are jointly applied, D^{2}FT-Net achieves the best performance with an mIoU of 73.49%, highlighting the complementary and synergistic effects of the two modules.

To validate the effectiveness of the AFA-Adapter, we conduct a comparative study with representative PEFT methods. As shown in Table VI, AFA-Adapter consistently outperforms

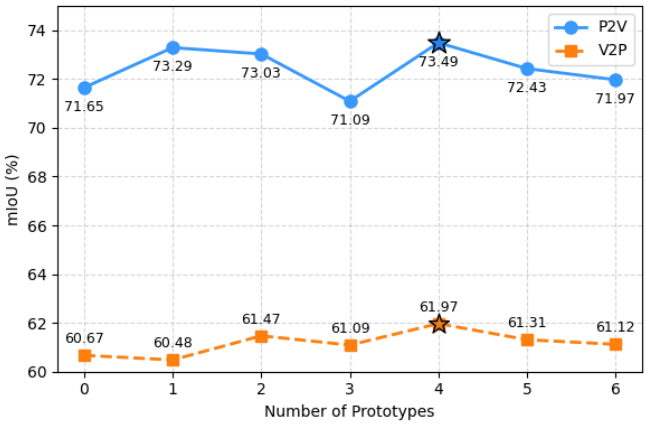


Fig. 7. Ablation study on the number of prototypes, reporting the mIoU performance on the ISPRS dataset.

VPT [50], AdaptFormer [51], and Earth-Adapter [8], demonstrating its superior effectiveness.

d) Hyperparameter-sensitive ablation: To investigate the sensitivity of D^{2}FT-Net to key hyperparameters, we conduct a series of ablation experiments by varying the number of prototypes used in the SMP-Adapter. Fig. 7 reports the corresponding mIoU performance under both P2V and V2P settings on the ISPRS dataset. As observed, a consistent trend is exhibited in both experimental settings: the segmentation performance initially improves as the number of prototypes increases, reaching the best results when four prototypes are adopted. This indicates that an appropriate number of prototypes is beneficial for modeling intraclass diversity across domains. However, further increasing the number of prototypes leads to performance degradation, suggesting that excessive prototypes may introduce redundancy and weaken cross-domain feature consistency. Overall, these results demonstrate that D^{2}FT-Net is relatively robust to the choice of prototype number within a reasonable range, and a moderate prototype setting provides a favorable balance between representation capacity and generalization ability.

4) Feature Analysis:

a) Feature comparison and frequency analysis: To intuitively investigate the working mechanism of AFA-Adapter, we further present CAM visualizations on representative categories, including building and tree, which exhibit significant cross-domain appearance variations. For comparison, the representative method Rein [6] is also evaluated in this experiment.

As shown in Fig. 8(a), for the tree category, Rein produces almost no effective response in the highlighted region, leading to incorrect predictions. In contrast, D^{2}FT-Net exhibits stronger and more concentrated activations, resulting in accurate segmentation. This difference mainly stems from the significant appearance discrepancies of trees across domains. However, after mapping the samples into the frequency domain and performing low-frequency reconstruction, both domains exhibit similar large-scale structural patterns, indicating that low-frequency components capture more stable and domain-invariant global characteristics. A similar phenomenon can

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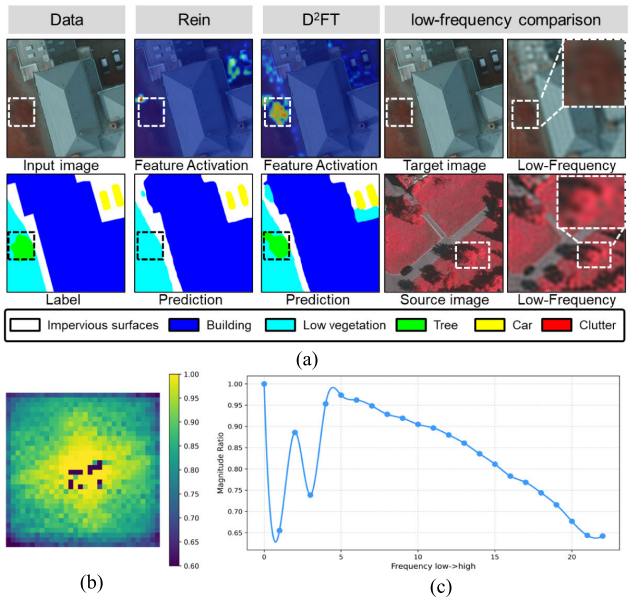


Fig. 8. CAM visualizations of representative tree-category scenarios, along with frequency analysis, comparing Rein with D²FT-Net. (a) CAM visualizations for the Tree and the low-frequency reconstructions in both domains. (b) Magnitude ratio map. (c) Radial magnitude ratio curve.

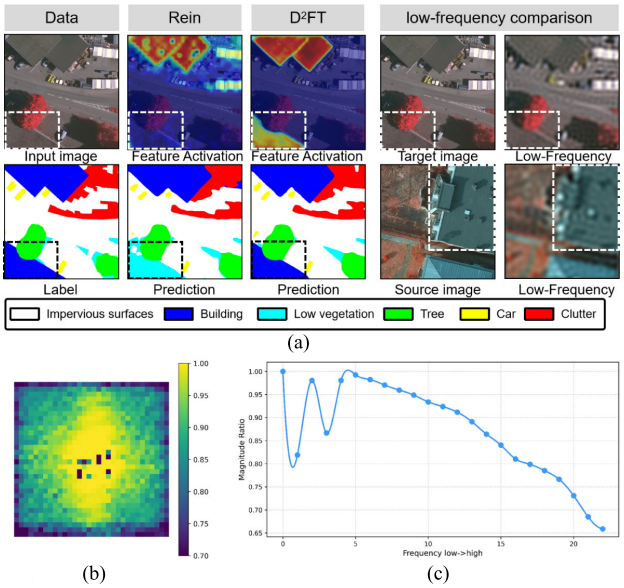


Fig. 9. CAM visualizations of representative building-category scenarios, along with frequency analysis, comparing Rein with D²FT-Net. (a) CAM visualizations for the Building and the low-frequency reconstructions in both domains. (b) Magnitude ratio map. (c) Radial magnitude ratio curve.

be observed for the building category in Fig. 9(a), where D²FT-Net generates more accurate responses than Rein. These observations indicate that, although there exist significant discrepancies between the source and target domains in the spatial domain, they exhibit high consistency in the low-frequency range, implying relatively small cross-domain intraclass variations.

Furthermore, as shown in Figs. 8(b) and 9(b), we visualize the magnitude ratio map, where the magnitude ratio is defined as the ratio of spectral magnitudes before and after the

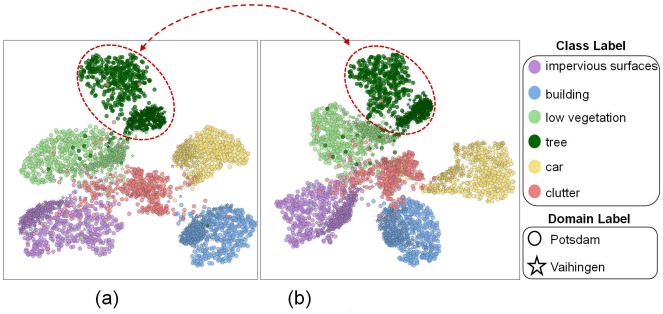


Fig. 10. UMAP visualization of interdomain feature clustering. (a) Rein. (b) Proposed D²FT-Net.

AFA-Adapter modulation, characterizing the relative preservation and enhancement of different frequency components during modulation. It can be observed that the magnitude ratio is generally higher near the spectrum center (corresponding to low-frequency regions) and lower in peripheral high-frequency regions, indicating that the proposed frequency modulation strategy tends to preserve and enhance cross-domain stable low-frequency information while suppressing high-frequency components that are more sensitive to domain variations.

In addition, Figs. 8(c) and 9(c) present the radial statistics of the magnitude ratio across different frequency radii, which quantitatively confirms this trend. As the frequency radius increases, the magnitude ratio gradually decreases, implying that low-frequency components are more strongly preserved, whereas high-frequency components are more effectively suppressed.

Overall, the AFA-Adapter adaptively selects low frequencies to effectively reduce the interference caused by spatial appearance discrepancies, thereby promoting VFMs to learn more consistent intraclass features across domains and, to a certain extent, alleviating large intraclass variations in cross-domain scenarios.

b) Feature distribution analysis: As shown in Fig. 10, we employ the UMAP technique to visualize the feature distributions extracted from two different domains, where samples from each domain are represented by dots and stars, respectively. Fig. 10(a) presents the features extracted by the Rein method, while Fig. 10(b) shows those obtained by the proposed D²FT-Net.

Compared with Rein, the features learned by D²FT-Net exhibit noticeably smaller intraclass distances across domains, with samples of the same category from different domains clustering more closely in the feature space. This observation indicates that D²FT-Net effectively alleviates the large cross-domain intraclass discrepancy by learning more compact and domain-consistent features.

D. Experiments on the LoveDA Dataset

To evaluate cross geographic domain generalization on the LoveDA dataset, we conduct experiments under two symmetric settings: urban-to-rural (U2R) and rural-to-urban (R2U). In U2R, the model is trained on the Urban training set and evaluated on the Rural test set, while in R2U, it is trained on

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TABLE VII

QUANTITATIVE COMPARISON WITH REPRESENTATIVE DG METHODS UNDER THE U2R SETTING SOURCE-ONLY DENOTES MODEL TRAINED ON THE SOURCE DOMAIN AND EVALUATED ON THE TARGET DOMAIN WITHOUT ANY GENERALIZATION ORACLE DENOTES MODEL TRAINED AND EVALUATED ON THE TARGET DOMAIN BOLD NUMBERS INDICATE THE BEST PERFORMANCE, AND UNDERLINED NUMBERS DENOTE THE SECOND-BEST RESULTS

Method	Backbone	Publication	Classes							mIoU (%)
			Background	Building	Road	Water	Barren	Forest	Agriculture	
DPCL [14]	ResNet-50 [47]	AAAI 2023	34.50	46.70	48.60	57.80	13.00	37.40	34.00	38.85
BlindNet [15]		CVPR 2024	55.20	37.00	40.80	52.00	5.10	27.00	48.50	37.98
HGFormer [19]	Swin-T [48]	CVPR 2023	58.55	48.16	42.10	67.41	10.64	27.56	47.97	43.20
CMFormer [27]		AAAI 2024	57.02	53.23	34.69	63.43	8.82	20.60	43.98	40.25
Source-only	DINOv2 [21]	—	54.86	58.67	47.04	55.82	12.81	31.99	54.11	45.04
Oracle		—	58.64	60.87	43.9	69.28	20.85	25.46	66.04	49.29
Rein [6]		CVPR 2024	**58.51**	59.99	47.52	66.39	10.21	24.89	56.16	46.24
FADA [31]		NeurIPS 2024	56.98	60.06	50.43	65.19	10.92	21.38	57.81	46.11
SET [32]		ACM 2024	55.23	62.16	46.02	63.94	12.96	27.17	58.89	46.62
CMPCL [24]		T-ITS 2024	57.66	56.39	**51.85**	64.01	16.23	19.75	51.97	45.41
GeoSA-BaSA [9]		ISPRS 2025	35.02	**62.70**	50.33	57.06	6.44	**37.88**	41.33	35.99
Earth-Adapter [8]		AAAI 2026	53.85	57.32	45.67	**68.12**	14.02	24.93	**63.48**	46.77
D^{2}FT-Net (Proposed)		—	56.70	60.69	49.89	64.33	**17.02**	30.82	59.45	**48.41** (+1.64)

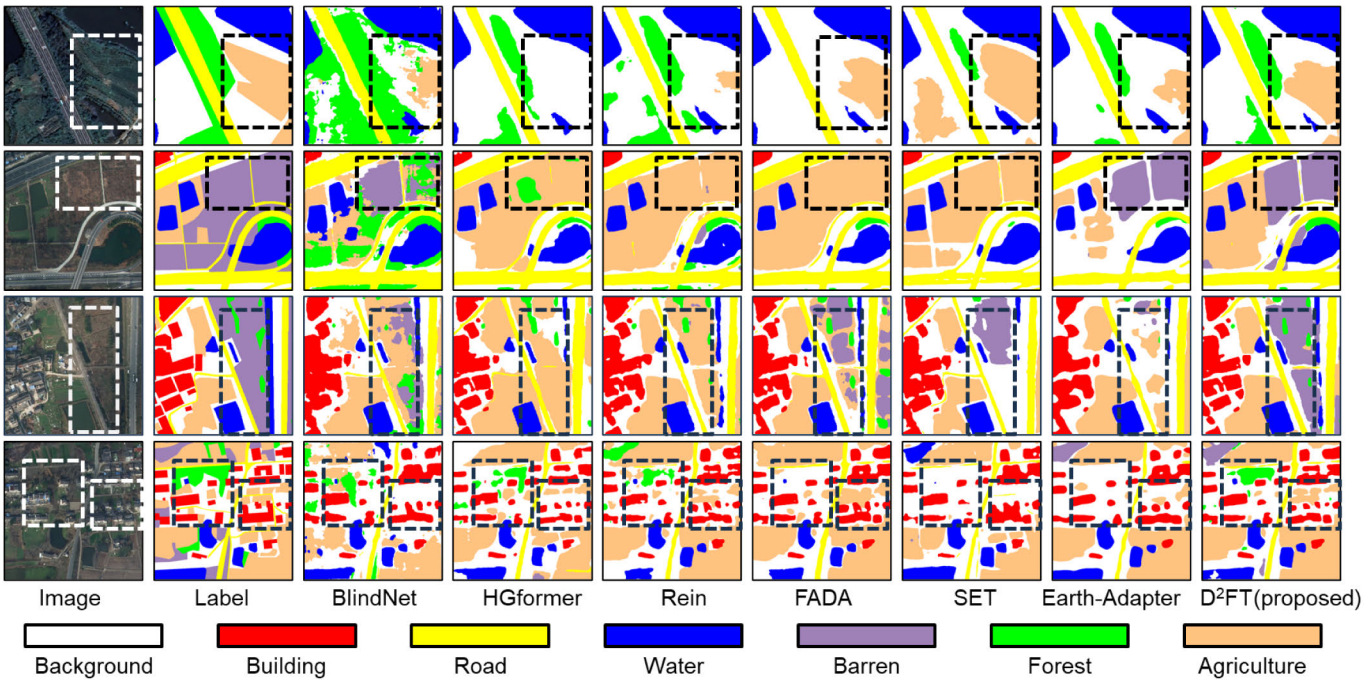


Fig. 11. Visualization results of the proposed D^{2}FT-Net and representative comparison methods on the U2R setting.

the rural training set and evaluated on the urban test set. In both settings, no target-domain annotations are used during training, ensuring a fair evaluation of cross-geographic generalization.

1) Urban-to-Rural:

a) Performance comparison: In this section, comprehensive experiments are conducted under the U2R cross-domain setting to quantitatively compare D^{2}FT-Net with other representative domain generalization methods. The experimental results of different methods are summarized in Table VII.

As shown in Table VII, the proposed method achieves an overall mIoU of 48.41%, which outperforms the SOTA method by 1.64% and yields the highest IoU in the barren category and ranks second in the agriculture category. This improvement can be attributed to the proposed method's ability to effectively reduce cross-domain intraclass discrepancies, leading to more consistent features across domains. In addition, the performance gains on these classes indicate that the proposed method is particularly effective in handling categories with large appearance variations, further demonstrating its robustness in cross-domain semantic segmentation scenarios.

b) Visualization of the proposed method: To provide a qualitative analysis of the segmentation performance,

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TABLE VIII

QUANTITATIVE COMPARISON WITH REPRESENTATIVE DG METHODS UNDER THE R2U SETTING SOURCE-ONLY DENOTES MODEL TRAINED ON THE SOURCE DOMAIN AND EVALUATED ON THE TARGET DOMAIN WITHOUT ANY GENERALIZATION ORACLE DENOTES MODEL TRAINED AND EVALUATED ON THE TARGET DOMAIN BOLD NUMBERS INDICATE THE BEST PERFORMANCE, AND UNDERLINED NUMBERS DENOTE THE SECOND-BEST RESULTS

Method	Backbone	Publication	Classes							mIoU (%)
			Background	Building	Road	Water	Barren	Forest	Agriculture	
DPCL [14]	ResNet-50 [47]	AAAI 2023	36.30	49.10	51.60	60.30	17.90	38.20	45.00	42.63
BlindNet [15]		CVPR 2024	35.50	44.50	48.90	54.60	30.10	35.50	45.30	42.05
HGFormer [19]	Swin-T [48]	CVPR 2023	42.57	60.72	58.07	73.51	51.35	55.40	59.12	57.25
CMFormer [27]		AAAI 2024	42.85	60.09	56.52	74.68	47.31	54.66	58.55	56.38
Source-only	DINOv2 [21]	—	43	62.31	60.1	74.41	46.94	51.27	60.12	56.88
Oracle		—	48.4	66.47	65.74	73.42	41.15	50.37	62.07	58.23
Rein [6]		CVPR 2024	40.93	63.79	**61.09**	74.57	51.89	55.16	61.92	58.48
FADA [31]		NeurIPS 2024	**43.98**	63.41	59.65	75.61	53.44	54.35	61.80	58.89
SET [32]		ACM 2024	43.25	65.07	59.04	75.33	49.97	54.86	61.76	58.47
CMPCL [24]		T-ITS 2024	42.48	**65.67**	55.12	**75.72**	49.08	56.61	**63.09**	58.25
GeoSA-BaSA [9]		ISPRS 2025	38.78	55.18	54.59	60.33	40.08	52.53	45.72	49.60
Earth-Adapter [8]		AAAI 2026	42.07	49.41	58.7	75.09	**54.08**	49.05	62.01	57.29
D²FT-Net (Proposed)		—	43.49	64.79	57.92	75.75	52.15	**57.07**	58.34	**59.28** (+0.39)

several representative scenes are selected for visual comparison. Fig. 11 illustrates the qualitative results of D²FT-Net and other representative methods under the U2R cross-domain setting. From left to right, the figure shows the input images, labels, and the predictions generated by different methods.

By comparing the key regions highlighted by the dashed boxes in each row, further differences among methods under complex cross-domain scenarios can be observed. In the first row, compared with other methods, the proposed approach more accurately segments the agricultural regions and effectively alleviates the misclassification of Background as Water, producing predictions that are more consistent with the label. In the second row, our method distinguishes the agriculture class more reliably than competing approaches. In the third row, the proposed method more completely identifies barren areas and correctly recognizes the forest targets within the barren regions. In the fourth row, our method achieves more accurate segmentation of the building and the surrounding forest regions, yielding clearer boundaries and more coherent structures. Overall, these qualitative results further demonstrate the effectiveness and robustness of the proposed method in challenging cross-domain scenarios.

2) Rural-to-Urban:

a) Performance comparison: In this section, systematic experiments are conducted under the R2U cross-domain setting to quantitatively evaluate the performance of D²FT-Net against other representative domain generalization methods. The experimental results of different methods are summarized in Table VIII.

As shown in Table VIII, the proposed D²FT-Net achieves an overall mIoU of 59.28%, surpassing the SOTA method by 0.39% mIoU. It also ranks among the top two results on the background, building, and forest classes, further validating its robust generalization capability under cross-domain scenarios. It is worth noting that the source-only model outperforms the oracle on water and forest categories due to the richer sample distribution of these classes in the rural domain, despite lagging in overall mIoU. Leveraging effective domain generalization fine-tuning, D²FT-Net surpasses both baselines on these categories while attaining the best overall performance.

b) Visualization of the proposed method: To provide a qualitative analysis of the segmentation performance, several representative scenes are selected for visual comparison. Fig. 12 illustrates the qualitative results of D²FT-Net and other representative methods under the R2U cross-domain setting. From left to right, the figure shows the input images, labels, and the predictions generated by different methods.

By comparing the key regions highlighted by the dashed boxes in each row, further differences among methods under complex cross-domain scenarios can be observed. In the first row, compared with other competing approaches, the proposed method produces segmentation results for forest and agriculture that are more consistent with the label. In the second row, the other methods fail to correctly recognize the road class, whereas our method accurately predicts this category. In the third row, the proposed method distinguishes Forest from agriculture more accurately than other methods. In the fourth row, compared with the competing approaches, our method yields clearer and more complete building boundaries, further demonstrating superior cross-domain robustness and spatial consistency.

3) Hyperparameter-Sensitive Ablation: To investigate the sensitivity of D²FT-Net to key hyperparameters, we conduct a series of ablation experiments by varying the number of prototypes used in the SMP-Adapter. Fig. 13 reports the corresponding mIoU performance on the LoveDA dataset under both U2R and R2U settings. As observed, both settings exhibit a consistent trend: the segmentation performance initially improves as the number of prototypes increases, reaching the best results when four prototypes are adopted, and then

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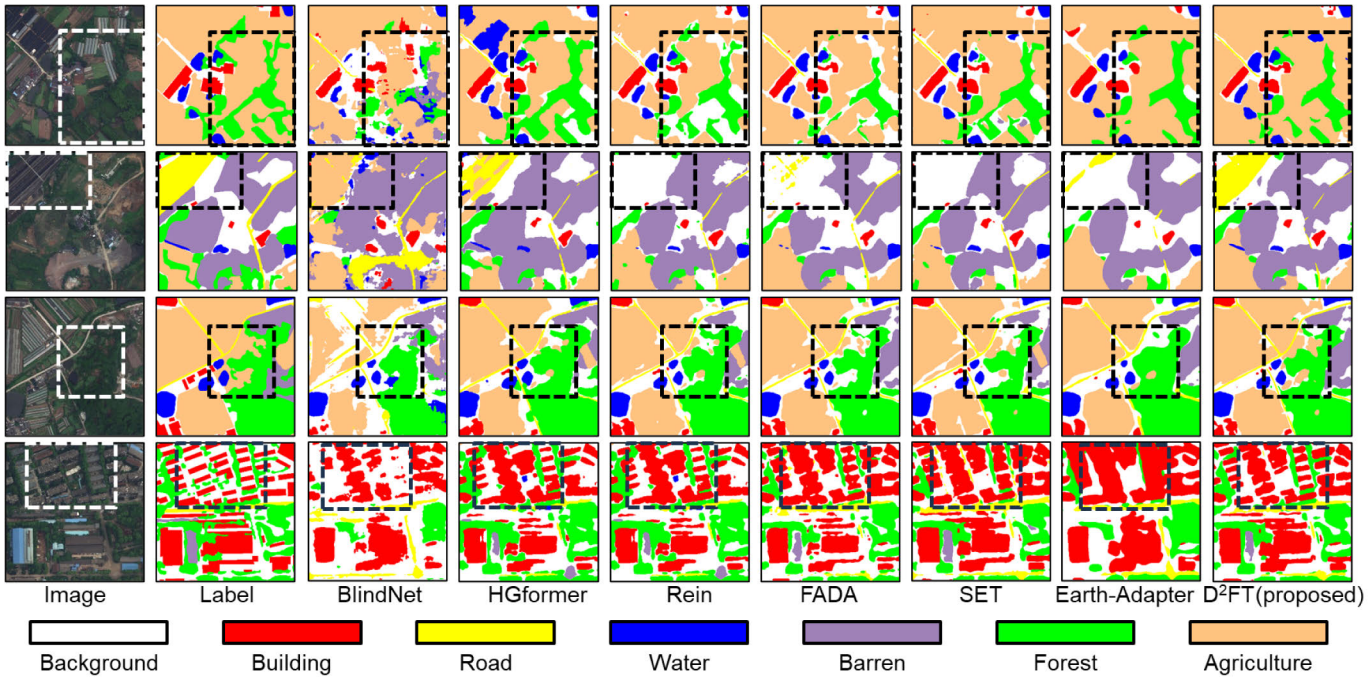


Fig. 12. Visualization results of the proposed D^{2}FT-Net and representative comparison methods on the R2U setting.

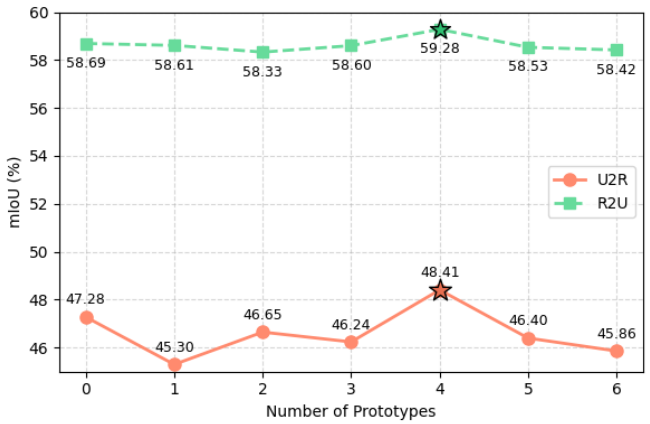


Fig. 13. Ablation study on the number of prototypes, reporting the mIoU performance on the LoveDA dataset.

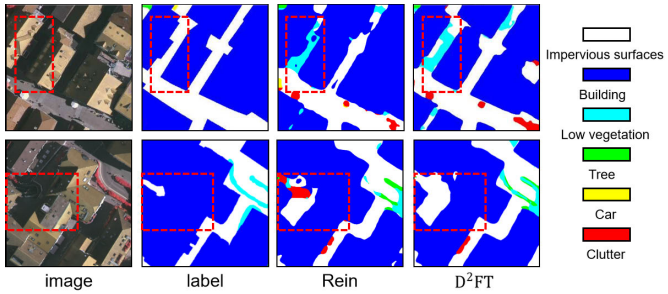


Fig. 14. Examples of failure cases. D^{2}FT-Net shows degraded performance in regions occluded by shadows cast by objects.

gradually declines as the number of prototypes continues to grow. This indicates that an appropriate number of prototypes is beneficial for effectively modeling intraclass diversity across domains.

E. Failure Cases

Despite the effectiveness of the proposed D^{2}FT-Net, certain challenging scenarios remain unresolved. As shown in Fig. 14, two representative failure cases are presented. For each case, from left to right, the columns correspond to the image, the label, the prediction of the representative method Rein [6], and the prediction of D^{2}FT-Net. Key regions are highlighted with red dashed boxes.

From these two cases, it can be observed that both Rein [6] and D^{2}FT-Net exhibit noticeable performance degradation in regions occluded by building shadows, resulting in significant misclassification. This indicates that under strong shadow interference, existing methods generally struggle to extract discriminative features, which also suggests potential directions for future improvements.

V. Conclusion

In this study, D^{2}FT-Net is proposed for remote sensing domain generalization semantic segmentation, aiming to address severe cross-domain intraclass variations. D^{2}FT-Net consists of two core components: AFA-Adapter and SMP-Adapter. The AFA-Adapter aims to learn cross-domain intraclass consistent feature clusters, while the SMP-Adapter aims to enhance the discriminability of easily confused features at feature cluster boundaries. By jointly adapting VFMs in both the frequency and spatial domains, D^{2}FT-Net alleviates cross-domain intraclass variations. Extensive quantitative experiments on two benchmark datasets under four representative cross-domain settings demonstrate the effectiveness and competitiveness of the proposed D^{2}FT-Net. Moreover, comprehensive feature analysis consistently confirms that D^{2}FT-Net

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effectively alleviates cross-domain intraclass variations in remote sensing imagery.

Based on the failure cases observed in the experimental results, the limitations of the proposed D²FT-Net are further analyzed. The model shows degraded performance in regions affected by building shadows, where significant appearance variations introduce feature-level semantic ambiguity and hinder accurate segmentation. Future work will focus on improving the robustness of the model under shadow-affected conditions.

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